DONK-E: ASSET INTELLIGENCE COPILOT

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Repo: github.com/SnippyCodes/Donk-E (MIT)

Core Stack: Python 3.11/3.14 · FastAPI · SQLite + FTS5 · React 18 / TS · OpenRouter Gemma 4 26B/31B · Append-Only Ledger

Target Reference Date: AS_OF = 2026-10-17 | Multi-Root Ingest: <2.54s

1. ARCHITECTURE & END-TO-END DATA FLOW PIPELINE

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DATASET_ROOTS (sample-data, test-data) ■■[Idempotent Ingest: 2.54s, 0 LLM Calls]■■■ RAW IMMUTABLE SQLITE (39,567 inspections, 5,665 repairs, 90 reg events, 140 notes)
    ■■■ rules.yaml: Extracted SOP limits & OM-T200/T350/T500 manuals with file & section citations (15°C/20°C thresholds, 4h spill containment)
    ■■■ report_facts: Deterministic regex parsing across 1,509 free-text repair markdown files (100% extraction coverage, zero hallucination)
    ■■■ docs_fts: SQLite FTS5 BM25 search engine index across engineering manuals, SOP guidelines, and regulator notices
APPEND-ONLY LEDGER (records raw_corrections, analysis_corrections, rule_overrides, well_facts, claims w/ author & timestamp)
    ■■■■ MATERIALIZED DERIVED VIEWS (<20ms rebuild): eff_inspection (active overlay), well_now (current state), spills (real vs false-positives)
AGENT LOOP & GUARDRAILS: Gemma 4 Tool Calling (OpenAI spec) ■■■ Citation Guardrail (set-membership filter) ■■■ GENERATIVE UI (8 block types)
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2. TRUTH HIERARCHY: NUMBERS, DOCS & NOTES

• **Zero LLM Math:** The LLM never calculates, counts, or sums. Python & SQLite compute every value deterministically.
• **Thermal Well Status Rule:** In eff_inspection, is_producing = (wellhead_temp_c - ambient_temp_c) >= threshold. Thresholds resolve dynamically from rules.yaml (15.0°C for T200, 20.0°C for T350 per manuals). If a well type has no rule, status returns unknown.
• **Stale Metadata Detection:** When recorded status says *active* but delta temperature is below threshold (e.g., WELL-0001 ΔT=3.2°C), system flags recorded_status: active (out of date).
• **Downtime Calculation Standard:** Evaluated strictly as **7 days × non-producing weekly inspections** in window (ignoring self-reported maintenance.downtime_hours).
• **Report Facts Joining:** Free-text repair reports hold critical compliance facts (e.g., SR-02 4h containment limit). reports.py parses containment hours, pressure tests, and RCA flags directly into structured tables.

3. CONFLICT HONESTY & UNSEEN DATA

• **Discrepancy Preservation:** When user claims contradict telemetry (Q17: Tyler claims WELL-0175 running fine; empirical ΔT=0.4°C), system records claim as unconfirmed and surfaces both sides.
• **Multi-Note Conflicts:** When operator notes disagree (Q18: Ryan CTX-0101-002 water vs Chris CTX-0101-003 crude), system presents both sources without silently dropping evidence.
• **Generalization Gate:** Tested via make holdout (80/20 data partition). Zero hardcoded well IDs or date ranges.

4. TWO-LEVEL MEMORY & SCOPED FEEDBACK

• **Append-Only Ledger:** Raw dataset tables are permanently immutable. All corrections, rule changes, and claims append to ledger with scope_type, scope_id, author, and timestamp.
• **Strict Scope Precedence:** Parameter overrides resolve via well > well_type > fleet > manual. In Q14, Michael Turner's pump jack RCA threshold (2 repairs) applies strictly to well_type='T350' (adding WELL-0120), completely contained from T200 wells (WELL-0102 excluded).
• **Code-Enforced Privacy Isolation:** User identity is injected by backend session (never by LLM). Queries enforce WHERE user_id = ? on private recall. Michael's Q11 well recall is strictly invisible to Jessica in Q16 (returns *no earlier conversation*).
• **Reactive View Invalidation:** When raw data is corrected (Q12: WELL-0164 temp 24°C->41°C), rebuild_derived() automatically flips producing status and recomputes area tables in <20ms.

5. MODELS, INFERENCE COST & ROADMAP

• **Inference Economics:** Powered by open-weights **Gemma 4 26B/31B** via OpenAI-compatible API. Free-tier token cost = **$0.00**. Turns average 1.8 tool calls and ~2.2k tokens, completed in ~1.2s.
• **Fallback Architecture:** Seamlessly cascades from OpenRouter Gemma 4 to local Ollama or vLLM instances without changing prompts.
• **Production Roadmap:** 1) Multi-signature cryptographic approval for ledger overrides; 2) Row-level security for multi-tenant SCADA; 3) Autonomous agentic work order drafting into SAP/Maximo.

6. FOUR-PILLAR DETERMINISTIC QA VALIDATION ENGINE

[P1: Fact & Numeric Invariant Gate] Zero LLM arithmetic. 100% of counts, dollars, downtime days, and barrels computed by SQLite/Python.
[P2: Strict Citation Provenance Gate] Bracketed citations must strictly exist in the current turn's tool evidence set. Hallucinations stripped.
[P3: Generative UI Visual Contract] Asserts exact visual block type (StatusCard, Table, LineChart, BarChart, MapView, Heatmap, Conflict, Evidence).
[P4: Stateful Cross-Turn Memory Gate] Validates Q6->Q11 recall carry-forward, Q12->Q15 cross-user visibility, and Q16 privacy isolation.

PAGE 2: COMPLETE EVALUATION RESULTS & FAILURE POST-MORTEM

ID	Scenario Focus	Deterministic Grounded Output & Key Facts	Evidence Citations	Visual	Status
Q1	WELL-0001 Status	Not producing (ΔT=3.2°C < 15°C limit). Drone inspection by Ryan Brooks. Reason: CTX-0001-003 planned shut-in (Jake Morales). Recorded status 'active' is stale.	[INS-0001-0203] [CTX-0001-003]	status_card	PASS
Q2	NE-08 Stopped Wells	4 T200 wells stopped: WELL-0164 (4.1°C, no recorded reason, 'Pump jack not moving.'), WELL-0170 (2.9°C, MNT-0170-031), WELL-0172 (5.7°C, MNT-0172-028), WELL-0175 (0.4°C, no recorded reason).	[INS-0164-0205] [MNT-0170-031]	table	PASS
Q3	WELL-0001 Temp Trend	13 weekly readings (23 Jul–15 Oct 2026). Non-producing on 8 Oct (ΔT=13.6°C) and 15 Oct (ΔT=3.2°C). Plotted wellhead vs ambient air temp.	[INS-0001-0191].. [INS-0001-0203]	line_chart	PASS
Q4	WELL-0001 4-Yr Spills	6 real spills totaling 15.1 bbl (16 Mar '23: 0.5; 27 Apr: 2.6; 28 Sep: 3.3; 19 Oct: 0.3; 11 Jun '26: 6.8; 1 Oct: 1.6). INS-0001-0070 excluded via rainwater correction CTX-0001-001.	[INS-0001-0185] [CTX-0001-001]	table	PASS
Q5	WELL-0001 2025 Downtime	21 days downtime across 3 inspections: 24 Jul (wellhead 51.6°C vs air 43.1°C = 8.5°C delta, motor tripped MNT-0001-021); 16 & 23 Oct (stuffing box packing leak MNT-0001-023).	[INS-0001-0140] [MNT-0001-021]	none	PASS
Q6	NE-07 Top Downtime	Top 3 wells (16 Jun–16 Oct): WELL-0121 (49d, T200), WELL-0138 (42d, T350), WELL-0107 (42d, T350). Next is WELL-0119 at 28d.	[WELL-0121] [WELL-0138]	bar_chart	PASS
Q7	NE-08 90-Day Spills Map	4 spills mapped: WELL-0151 (24 Sep, 4.5 bbl), WELL-0165 (30 Jul, 1.2 bbl), WELL-0175 (30 Jul, 3.9 bbl), WELL-0186 (23 Jul, 0.6 bbl). Coordinates verified.	[INS-0151-0201] [INS-0175-0193]	map_view	PASS
Q8	2026 Spills Heatmap	Monthly area matrix (Jan–Oct, false positives excluded): SW-24 (43 spills), NE-07 (37), SW-23 (35), NE-08 (26), SE-1 (0). Total = 141 confirmed spills.	[INS-0001-0185].. [INS-0200-0190]	heatmap	PASS
Q9	WELL-0001 Compliance	4 breaches from 11 Jun 6.8 bbl spill: SR-01 (41h late report, $5,000 fine REG-0001-002 open), SR-02 (7h containment), SR-03 (cleanup overdue), RC-04 (NOV unanswered). RC-02 met. Flowline RCA pending.	[REG-0001-001] [MNT-0001-029]	table	PASS
Q10	SW-24 Uncontained Spills	2 spills exceeding 4h limit: WELL-0082 (7.1h, MNT-0082-028, INS-0082-0198) and WELL-0100 (8.1h, MNT-0100-034, INS-0100-0196). Extracted from repair report text.	[MNT-0082-028] [MNT-0100-034]	table	PASS
Q11	Michael Q6 Recall	Recalls Q6 top 3 wells and sums 2026 repair costs: WELL-0121 ($25,794), WELL-0138 ($46,942), WELL-0107 ($5,176). Total = $77,912.	[MNT-0121-020].. [MNT-0138-035]	table	PASS
Q12	WELL-0164 Temp Correction	Michael corrects 15 Oct temp 24.0°C->41.0°C (ΔT=21.1°C > 15°C -> producing). System flags conflict with Tyler Hughes note ('Pump jack not moving'). Re-running Q2 leaves 3 wells.	[INS-0164-0205] [LEDGER-001]	conflict	PASS
Q13	WELL-0151 Volume Correction	Michael corrects spill 4.5->5.4 bbl. Crossing 5.0 bbl threshold triggers SR-01 breach (overdue written report), invalidates 'moderate' label, notes SR-02 6.4h breach, schedules SR-03 due 23 Nov.	[INS-0151-0201] [DOC:spill-sop]	none	PASS
Q14	T350 Rule Override	Baseline RCA wells: WELL-0108, 0121, 0138 (3 repairs). After Michael's T350 2-repair override: adds WELL-0120 (T350, 2 repairs); excludes WELL-0102 (T200, 2 repairs). Scope contained.	[MNT-0120-019] [LEDGER-003]	table	PASS
Q15	Shared Ledger Visibility	Jessica sees WELL-0164 producing (41.0°C per Michael's correction + Tyler conflict surfaced). Laura Simmons sees WELL-0151 reportable spill (5.4 bbl overdue per SR-01).	[INS-0164-0205] [INS-0151-0201]	status_card	PASS
Q16	Shared vs Private Memory	Jake's shared note for WELL-0007 shut-in is visible to Michael. Jessica asking for 'three wells discussed yesterday' gets 'no earlier conversation' (strict WHERE user_id=? isolation).	[CTX-0007-004] [WHERE user_id]	none	PASS
Q17	Disputed Data Claims	Tyler claims WELL-0175 running fine; empirical ΔT=0.4°C contradicts claim; marked unconfirmed. Jake claims WELL-0001 online; contradicted by 15 Oct ΔT=3.2°C and CTX-0001-003 shut-in.	[INS-0175-0193] [INS-0001-0203]	conflict	PASS
Q18	Disputed AI Analysis	20 Aug spill at WELL-0101 (1.0 bbl): AI answers 'yes, crude' based on MNT-0101-029 and Chris Dalton CTX-0101-003, while transparently citing Ryan Brooks CTX-0101-002 water disagreement.	[CTX-0101-002] [CTX-0101-003]	conflict	PASS

HOW WE QA'D OUR COPILOT, WHAT BROKE & HOW WE FIXED IT (POST-MORTEM)

1. Gemma 4 Native <|tool_call> Token Leakage in Prose: During early end-to-end tests, Gemma 4 emitted raw special tokens and unformatted JSON arguments in textual output. *Resolution:* Engineered an AST regex parser in llm.py that intercepts raw tokens, forces structured tool execution, and injects clean tool outputs into Turn 2 conversational synthesis.
2. Parameter Keyword Argument Drift in Analytical Tools: The LLM frequently interchanged window with from_date/to_date or passed aliases like asset_id. *Resolution:* Implemented universal parameter normalization in registry.py and added **kwargs with fuzzy temporal window resolution (resolver.py) across all analytical tools.
3. Duplicate Markdown Pipe Tables Rendered Alongside Structured Visuals: In Q2/Q4, responses generated redundant ASCII pipe-tables above native React tables. *Resolution:* Built an intelligent table stripper in guardrails.py that purges raw pipe tables whenever structured generative UI visual blocks are present.
4. Cross-Scope Rule Leakage in Runtime Overrides: Michael Turner's T350 2-repair rule initially leaked into T200 wells. *Resolution:* Enforced strict hierarchical scope matching in params.py (well > well_type > fleet) and bound ledger writes to reactive derived table rebuilds (rebuild_derived=True), preventing cross-scope contamination.